

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	An answer that includes the following points: • description of the use of all three solutions to grow onions (1) • description of a relevant biotic factor kept constant (1) • description of a relevant abiotic factor kept constant (1) • appropriate measurement of {plant parts / plant} made at start and finish (1) • after {same/ specified / stated} time (1)	No mark if growing in soil e.g. age, variety (ignore species) NOT size as it is DV e.g., light/ temperature / pH of solution e.g. height, length, mass Accept if it is clear all started at same height/mass etc. / if measure difference or increase in height / mass etc. if time quoted must be a week minimum	expert (4)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	An answer that includes the following steps: • correct values from graph correctly manipulated (1) • correct percentage (1)	e.g. 5g for shoot + bulb, 6.2 g for total e.g. 80.6 (%) correct with no working gains both marks allow even if not in table if no working allow correct answer which is incorrectly rounded for 1 mark (80.64 or 81)	expert (2)

Question Number	Answer	Additional Guidance	Mark
3(a)(iii)	An answer that includes the following points: • as solution concentration goes up growth (of all parts) goes down (1) • the effect of solution concentration on mass is {greatest in shoots / least in roots} (1) • percentage edible is unaffected / little affected by solution concentration (1)	Accept higher / est (for all parts) in ½ strength Accept lower / est (for all parts) in 2x strength Accept reverse argument Allow description of relative effects, e.g. percentage increase in bulbs greatest ALLOW double strength solution reduces edible percentage or reverse Allow ecf if aii wrong	expert (3)

Question Number	Answer	Additional Guidance	Mark
3(a)(iv)	A description that includes three of the following points: - grow several plants in each solution (1) - calculation of (means and) SDs (1) - check for overlap (of means plus and minus SD) for each difference (1) - carry out a statistical test (1)	Accept repeat experiment ALLOW accept ranges, error bars in place of SD ALLOW t test, ignore Chi, correlation	expert (3)

Question Number	Answer	Additional Guidance	Mark
3(b)	An answer that includes the following points: - remove potassium hydrogen phosphate (1) - replace potassium removed / add extra of named salt to adjust balance (1)	Accept don't add KH_2PO_4 / use solution without KH_2PO_4 Ignore remove phosphate ions Accept idea that as there is KNO_3 there is no need to do anything but remove KH_2PO_4	Expert (2)

Question Number	Answer	Additional Guidance	Mark
3(c)	An answer that includes the following points: • because this is the highest (uptake) of all mineral ions • {below 6.5 / at 8.0-8.5} phosphate in short supply / {above 8.0 / below 5.5} nitrate in short supply (1) • needed for appropriate {substance / process} (1) • pH {above / below} this range will denature enzymes (1)	e.g ATP, DNA, RNA, nucleotides, cell division, energy exchange, protein synthesis. Allow ecf from mp2	expert (4)